TikTok Music Virality Analysis

Data Collection Progress | January 2026

Project Overview

- Objective: Analyze what makes songs go viral on TikTok
- Approach: Collect multi-source music data for analysis
- Goal: Build predictive models for song virality
- Focus: Audio features, lyrics, and platform-specific metrics
- Application: Help artists and labels optimize for TikTok success

Data Collection Strategy

- Multi-platform approach for comprehensive coverage
- TikTok trending tracks with Spotify audio features
- Spotify editorial playlists (TikTok Viral collections)
- Genius lyrics database for text analysis
- Video and creator metadata from TikTok
- Total: 490,000+ records across all sources

Datasets Collected

Dataset	Records	Key Features
TikTok Trending Tracks	6,747	Audio features, popularity, tempo
Spotify TikTok Viral	509	Track metadata, playlist source
Genius Lyrics	480,000+	Full lyrics, artist metadata
TikTok Videos	1,400+	Views, likes, shares, hashtags
Trending Authors	200+	Creator statistics

TikTok Trending Tracks Dataset

- 6,747 songs that trended on TikTok
- Linked with Spotify audio features:
  - Danceability, Energy, Valence (mood)
  - Tempo, Key, Loudness
  - Speechiness, Acousticness, Instrumentalness
- Includes popularity scores and release dates
- Perfect for ML feature engineering

Spotify TikTok Viral Playlists

- Scraped 3 major TikTok editorial playlists:
  - TikTok Music 2026 Viral Trending (201 tracks)
  - Viral TikTok Songs 2025 (131 tracks)
  - TikTok Hits 2024 (229 tracks)
- 509 unique tracks after deduplication
- Current/recent viral songs for validation
- Includes track popularity and release dates

Genius Lyrics Database

- 480,000+ English songs with full lyrics
- Source: HuggingFace (brunokreiner/genius-lyrics)
- Enables NLP and sentiment analysis
- Key fields: lyrics, artist, song title, metadata
- 302 MB dataset in Parquet format
- Can be matched with TikTok/Spotify data

Audio Features for Analysis

Rhythm & Energy

- Danceability: Suitable for dancing (0-1)
- Energy: Intensity and activity (0-1)
- Tempo: Beats per minute (BPM)
- Loudness: Overall volume (dB)

Mood & Character

- Valence: Musical positivity (0-1)
- Speechiness: Presence of words (0-1)
- Acousticness: Acoustic vs electronic
- Instrumentalness: Vocal presence

Challenges & Limitations

- TikTok API (Unofficial) - Bot detection blocked all requests
  - Tried TikTokApi library with multiple browser configurations
  - TikTok actively detects and blocks automated scraping
- Tokchart.com - JavaScript-rendered, requires headless browser
- TikTok Research API (Official) - Restricted access
  - Only available to academic researchers (PhD students)
  - Approval process takes approximately 4 weeks
- Workaround: Used Kaggle datasets + Spotify API successfully

Data Collection Pipeline

- Automated Python scripts for reproducibility
- Kaggle API for TikTok datasets
- Spotify API (spotipy) for playlist extraction
- HuggingFace datasets library for lyrics
- Data stored in CSV and Parquet formats
- Modular design for easy updates

Next Steps

Data Processing

- Clean and preprocess all datasets
- Match songs across platforms
- Engineer virality features
- Create train/test splits
- Handle missing values

Analysis & Modeling

- Exploratory data analysis (EDA)
- Correlation analysis
- NLP on lyrics (sentiment, topics)
- Build classification models
- Feature importance analysis

Summary

- Successfully collected 490,000+ music records
- Multi-source approach: TikTok, Spotify, Genius
- Rich audio features for ML modeling
- Lyrics data enables NLP analysis
- Pipeline ready for data processing phase
- Foundation set for virality prediction models